Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No : 92301733057**

Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Colab

Theory:

Linear regression is a method for determining the best linear relationship between two variables *X* and *Y*. If variables *X* and *Y* are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between *X* and *Y* then linear regression may be a useful means to describe the relationship between the two variables. The usual approach is to use the *least-squares* method, which minimizes the squared difference between the actual data points and a straight line. Let $[x_i, y_i]$, $i = 1,2,3,\ldots,N$ be the *N* pairs of data values of the variables *X* and *Y*. The straight-line relating *X* and *Y* is $y = mx + c$, where *m* and *c* are the gradient and constant values (to be determined) defining the straight line. Thus, $y(x_i) - y_i$ is the difference between the line and data point *i* (see Fig. 1). Taking all the data points, we seek values of *m* and *c* that minimize the squared difference *SD*.

$$\sum_{1}^{N} \left[y(x_i) - y_i \right]^2$$

This is achieved by calculating the partial derivatives of *SD* with respect to *m* and *c* and finding the pair [*m*,*c*] such that *SD* is at a minimum.

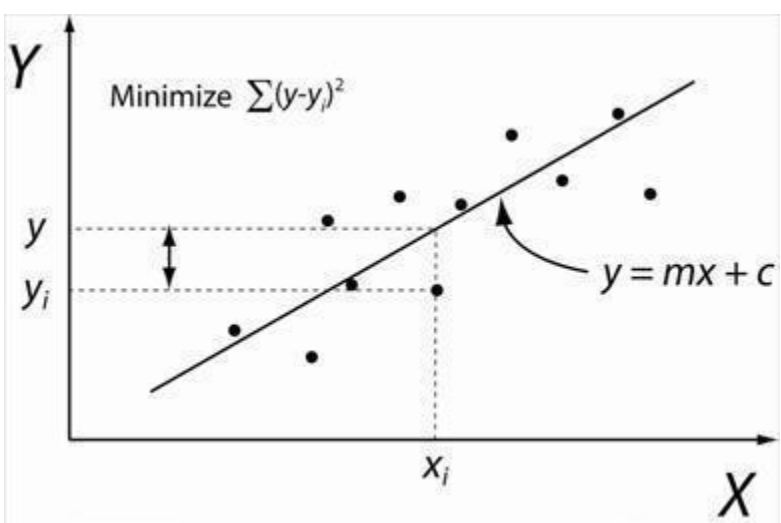

Figure 1: Illustration of Linear Regression. Linear least squares regression, the idea is to find the line y = mx + c that minimizes the mean squared difference between the line and the data points

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Batch Gradient Descent:

Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model. Batch gradient descent which processes all the training examples for each iteration of gradient descent. But if the number of training examples is large, then batch gradient descent is computationally very expensive.

Let m be the number of training examples. Let n be the number of features.

Algorithm for batch gradient descent :

Let $h_\theta(x)$ be the hypothesis for linear regression. Then, the cost function is given by:
Let Σ represents the sum of all training examples from i=1 to m.
$J_{train}(\theta) = (1/2m)\ \Sigma(\ h_\theta(x^{(i)})\ -\ y^{(i)})^2$

Repeat {
$\theta j = \theta j - (\text{learning rate}/m) * \Sigma(\ h_\theta(x^{(i)})\ -\ y^{(i)})x_j^{(i)}$
For every j =0 …n
}
Where $x_j^{(i)}$ Represents the jth feature of the ith training example. So if *m* is very large(e.g. 5 million training samples), then it takes hours or even days to converge to the global minimum.That's why for large datasets, it is not recommended to use batch gradient descent as it slows down the learning.

Pre Lab Exercise:

a. Explain the meaning of linear regression

- Linear Regression is a method used to predict a continuous value by finding a straight-line relationship between input and output.

b. Write three applications of linear regression
- Predicting house prices
- Sales forecasting
- Predicting student marks

c. Write three advantages of linear regression

- Simple and easy to understand
- Fast to train and implement
- Works well for linear relationships

d. Write three limitations of linear regression

- Assumes a linear relationship between variables
- Sensitive to outliers
- Not suitable for complex (non-linear) data

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Write the Hypothesis Function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Find the best fit line to the given dataset
12. Observe the cost function vs iterations learning curve

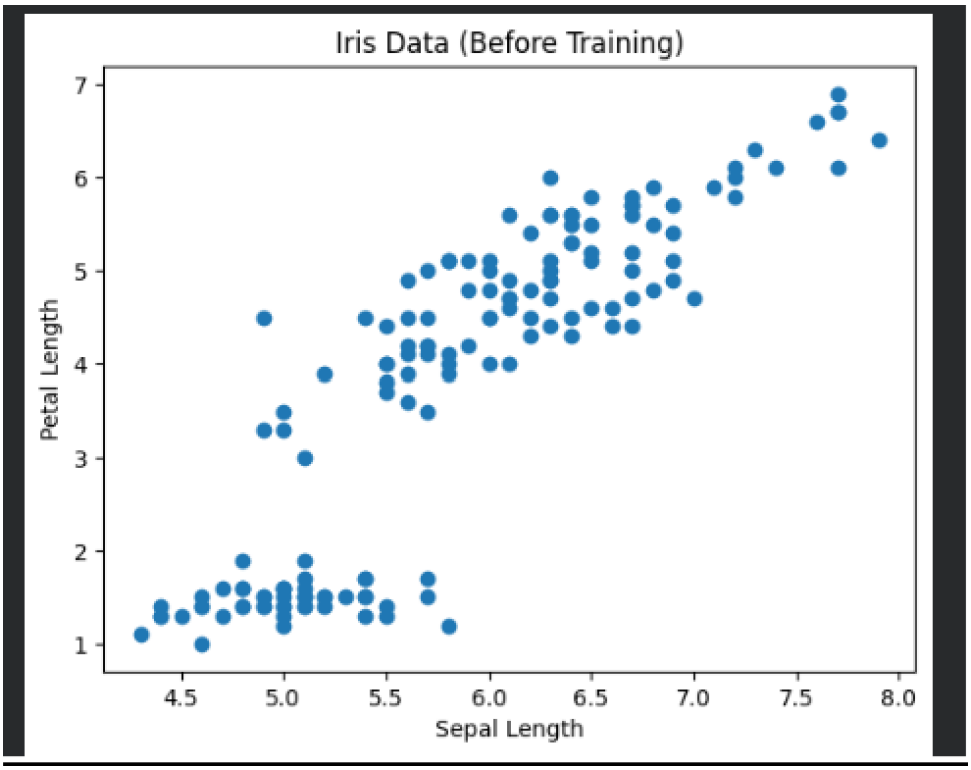
Program (Code):

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Take only ONE feature (as required in experiment)
X = df['sepal length (cm)'].values

# Target (you can choose any feature as output)
y = df['petal length (cm)'].values
```

```
# Scatter plot (before training)
plt.scatter(X, y)
plt.title("Iris Data (Before Training)")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()
m = len(X)

theta0 = 0  # intercept
theta1 = 0  # slope

learning_rate = 0.01
iterations = 1000

cost_history = []
def hypothesis(x, theta0, theta1):
    return theta0 + theta1 * x
def compute_cost(X, y, theta0, theta1):
    m = len(X)
    cost = 0

    for i in range(m):
        cost += (hypothesis(X[i], theta0, theta1) - y[i]) ** 2

    return cost / (2 * m)
for _ in range(iterations):
    temp0 = 0
    temp1 = 0

    for i in range(m):
        error = hypothesis(X[i], theta0, theta1) - y[i]
        temp0 += error
        temp1 += error * X[i]

    theta0 = theta0 - (learning_rate / m) * temp0
    theta1 = theta1 - (learning_rate / m) * temp1

    cost_history.append(compute_cost(X, y, theta0, theta1))
y_pred = theta0 + theta1 * X

plt.scatter(X, y, label="Actual Data")
plt.plot(X, y_pred, color='red', label="Best Fit Line")
```

```
plt.title("Best Fit Line (Iris Dataset)")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.legend()
plt.show()
plt.plot(range(iterations), cost_history)
plt.title("Cost Function vs Iterations")
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.show()
```

Results:



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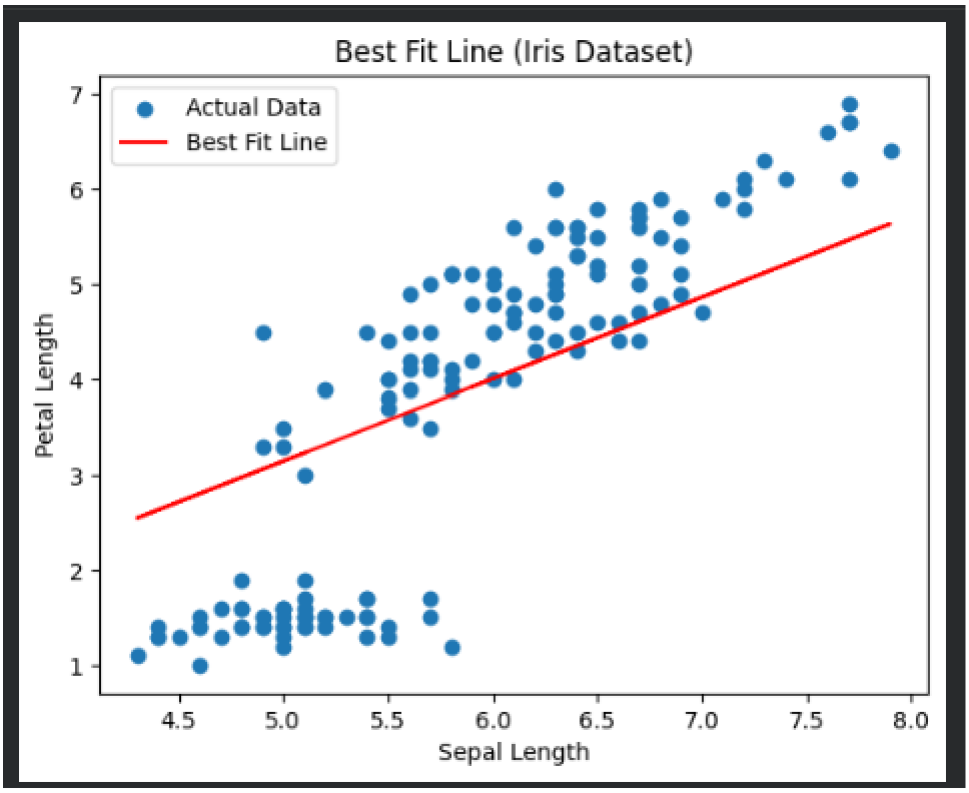

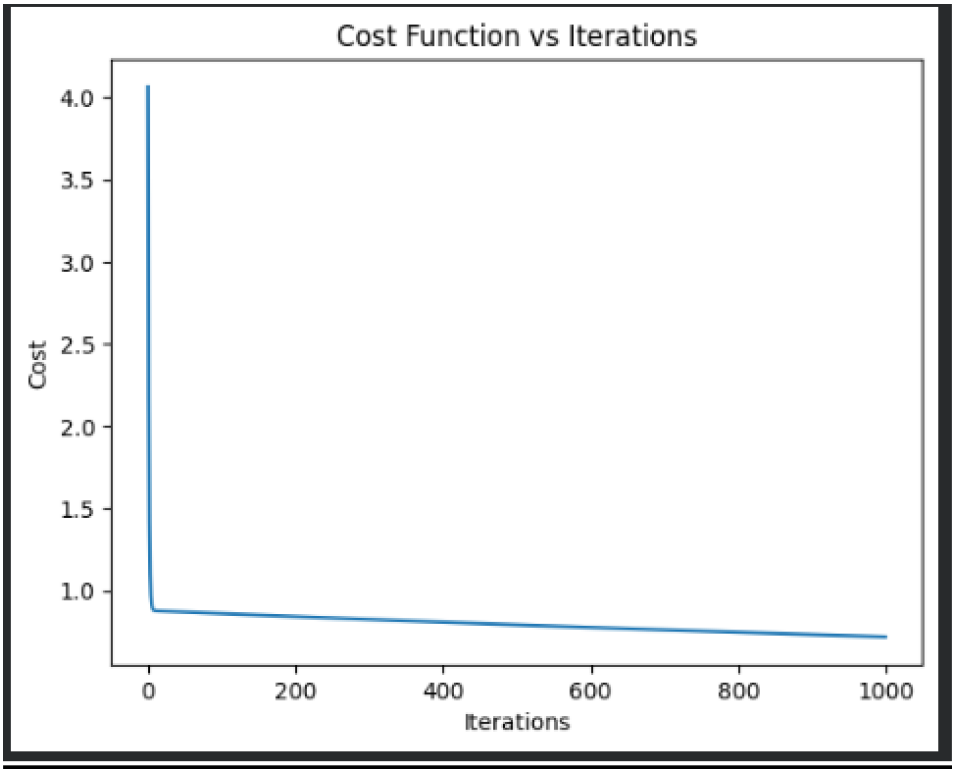

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Observation and Result Analysis:

a. Nature of the dataset

• The Iris dataset is a widely used dataset in machine learning, containing numerical features such as sepal length, sepal width, petal length, and petal width. In this experiment, sepal length is selected as the input variable, while petal length is used as the output variable.
• The dataset is clean and free from missing values. A roughly linear relationship can be observed between the chosen input and output features, making it appropriate for applying linear regression.

b. During Training Process

• During training, the model parameters (θ_0 and θ_1) are initialized and iteratively updated using the gradient descent algorithm. In each iteration, predicted values are compared with actual values to compute the error.
• The cost function (Mean Squared Error) is minimized gradually by adjusting parameters in the direction of the negative gradient. As iterations progress, the cost decreases steadily, indicating that the model is learning and improving its predictions.

c. After the training Process

- After training, the model identifies the best-fit line for the dataset. The final values of θ_0 and θ_1 represent the optimal parameters that minimize the difference between predicted and actual values.
- The scatter plot along with the regression line shows that most data points lie close to the line, suggesting a good fit. The cost function reaches its minimum value, confirming convergence.
- Thus, the model successfully captures the linear relationship between the input and output variables using linear regression.

Post Lab Exercise:

a. What are the major assumptions considered in linear regression
- Linearity – Relationship between variables is linear
- Independence – Observations are independent
- No Multicollinearity – Features are not highly correlated
- Normality of Errors – Errors follow normal distribution
- No Autocorrelation – No correlation between residuals
- No Measurement Error – Input variables are measured correctly

b. Why MSE is used instead of MAE for calculating the loss function
- MSE (Mean Squared Error) is differentiable, which makes it easier to optimize using gradient descent.

- It penalizes larger errors more due to squaring, helping the model reduce big mistakes.
- It provides a smooth gradient, leading to faster and stable convergence.

c. How can the behaviour of outliers be understood while dealing with the unseen dataset

- Outliers can be identified using IQR, box plots, or z-score methods.
- They may negatively affect model performance and predictions.
- Proper techniques like removal, transformation, or capping can be used to handle them.

d. Derive the Normal Equation for the Linear Regression.

The goal of linear regression is to minimize the cost function (Mean Squared Error):

$$J(\theta) = (X\theta - y)^T(X\theta - y)$$

To find the optimal parameters, we take the derivative of the cost function with respect to θ and set it to zero:

$$\frac{\partial J(\theta)}{\partial \theta} = 2X^T(X\theta - y) = 0$$

Solving for θ:

$$X^T X\theta = X^T y$$

$$\theta = (X^T X)^{-1} X^T y$$

$\theta = (X^T X)^{-1} X^T y$

This equation is known as the Normal Equation. It provides a direct solution to compute the optimal parameters θ without using iterative methods like gradient descent.

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